

# Feature Roadmap

Master list of candidate features, organized by role. Compiled from a full codebase review (June 2026). Effort tags: **S** = days, **M** = 1–2 weeks, **L** = multi-week.

[**GAP+**] = confirmed missing in code (not just assumed). (*infra: ...*) = reuses

packages/services already in the app.

## Cross-role / Platform

| #  | Feature                                               | Effort | Notes                                                                                 |
|----|-------------------------------------------------------|--------|---------------------------------------------------------------------------------------|
| 1  | UPI deep-link fee payment                             | M      | upi://pay intent via url_launcher; guardian claims -> staff confirms; no gateway fees |
| 2  | Automatic absence alert to guardians                  | S      | Cloud Function on attendance write -> FCM topic; feeds existing leave flow            |
| 3  | WhatsApp share button on every PDF                    | S      | share_plus already integrated; receipts, report cards, certificates                   |
| 4  | Admit card / hall ticket + datesheet PDF              | S      | pdf+printing infra proven by report cards                                             |
| 5  | Bonafide & Transfer Certificate generator             | S      | same PDF infra; pairs with student-exit workflow                                      |
| 6  | Student ID card generator (batch per class)           | S      | image_picker + pdf; replaces outside vendors                                          |
| 7  | Merit certificate generator                           | S      | leaderboards + PDF pipeline already exist                                             |
| 8  | Syllabus coverage tracker                             | M      | per class-subject chapter checklist; principal sees % covered                         |
| 9  | Study materials repository                            | M      | file_picker + Storage + compress already wired                                        |
| 10 | Transport module lite (routes, stops, driver contact) | M      | no GPS in v1; tap-to-call driver                                                      |
| 11 | Health records + incident log                         | M      | blood group, allergies; incident -> guardian push                                     |
| 12 | UDISE+ / govt report CSV export                       | S      | csv package present; enrollment by class/gender/category                              |
| 13 | Announcement read receipts ("seen by 23/40")          | S      | tiny seen write per open                                                              |
| 14 | Gate pass / early pickup with approval + push         | M      | safety feature; guardian request -> teacher approve -> gate pass                      |
| 15 | Parent polls & surveys with live tally                | S      | reuses announcement + notification plumbing                                           |
| 16 | Per-student performance trend charts                  | S      | fl_chart + marks across exams already stored                                          |
| 17 | Digital class diary (one line per class per day)      | S      | diary/{classKey}/{dateKey}; replaces paper diary                                      |
| 18 | Homework completion tracking (done / photo submit)    | M      | closes the homework loop; per-assignment teacher view                                 |
| 19 | SMS fallback for non-smartphone guardians             | M      | critical alerts via SMS gateway when no FCM token                                     |
| 20 | Academic year close / rollover wizard                 | M      | end-of-session: promote, archive, carry dues, reset counters                          |
| 21 | Online MCQ quizzes (auto-graded)                      | L      | teacher creates, students/guardians answer, auto marks                                |
| 22 | Library lite (book register, issue/return)            | M      | simple register, due-date reminders via existing push                                 |
| 23 | Event RSVP (sports day, annual function)              | S      | announcement + votes sub-map                                                          |
| 24 | Regional language expansion beyond EN/HI              | M      | i18n map pattern in lib/l10n/app_strings.dart scales                                  |

## Owner

| #  | Feature                                                             | Effort | Notes                                                                              |
|----|---------------------------------------------------------------------|--------|------------------------------------------------------------------------------------|
| 1  | Expense tracking -> real P&L                                        | M      | income side exists in fees; add expenses + monthly P&L PDF. Flagship owner feature |
| 2  | Daily cash reconciliation per collector                             | S      | grouping query over payments + "deposited" confirmation; anti-fraud                |
| 3  | Receipt void/edit protection (approval + report)                    | S      | targets the classic cancel-receipt fraud; audit_log exists                         |
| 4  | Carry-forward dues at promotion [ <b>verify promotion_service</b> ] | S      | unpaid dues must follow student into new session                                   |
| 5  | Projected cash flow (30/60/90-day expected collections)             | S      | derive from instalment due dates already stored                                    |
| 6  | Concession/discount approval workflow + annual report               | M      | who granted what to whom; second leakage point                                     |
| 7  | Teacher punctuality / accountability card                           | S      | aggregate leaves, substitutions caused, overdue tasks                              |
| 8  | Enrollment trend + dropout alerts with exit reasons                 | S      | wire deletion flow + fl_chart; "Class 7 lost 5 students"                           |
| 9  | Parent complaint inbox (assigned -> resolved, visible status)       | M      | reputation management; pattern detection                                           |
| 10 | Owner evening digest push (collection, attendance, admissions)      | S      | clone principal_digest pattern                                                     |
| 11 | Staff attendance + salary slip lite                                 | L      | check-in via offline-queue pattern; month-end payslip PDF                          |
| 12 | Admission inquiry tracker (CRM lite)                                | M      | inquiry -> visited -> admitted funnel; follow-up dates                             |
| 13 | Student document vault                                              | M      | Storage + DPDP consent infra make it defensible                                    |
| 14 | Inventory / asset register                                          | M      | benches, projectors, lab equipment; owner-facing                                   |
| 15 | School data backup/export (full CSV/JSON dump)                      | S      | owner peace-of-mind; vendor-lock-in antidote                                       |

## Principal

| #  | Feature                                                                       | Effort | Notes                                                                          |
|----|-------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------|
| 1  | Exam result analytics tab [ <b>GAP+ — analytics is attendance/fees only</b> ] | M      | pass %, subject averages, below-40% counts; 5th tab in analytics_screen        |
| 2  | Teacher-wise result comparison                                                | S      | join timetable (teacher->class->subject) to marks                              |
| 3  | At-risk student early warning                                                 | M      | fuse attendance drop + marks fall + remarks                                    |
| 4  | Classroom observation register (rubric + history)                             | M      | staff_remarks is free-text; appraisal season needs structure                   |
| 5  | Teacher compliance scorecard (per-teacher green/amber/red)                    | S      | attendance marked? homework posted? marks on time? from existing timestamps    |
| 6  | Teacher workload balance report                                               | S      | periods/week from timetable + substitution load                                |
| 7  | Question paper approval workflow                                              | M      | submit (file upload) -> approve/return -> lock; mirrors leave-approval pattern |
| 8  | Staff circular acknowledgment ("12 of 15 acknowledged")                       | S      | read-receipt mechanic on announcements, staff-flavored                         |
| 9  | Inspection pack (one-tap PDF bundle for DEO/board visits)                     | M      | enrollment, attendance, teacher list, meeting minutes                          |
| 10 | Lesson plan submission & weekly approval                                      | M      | pairs with syllabus tracker                                                    |

## Coordinator

| # | Feature                                    | Effort | Notes                                                          |
|---|--------------------------------------------|--------|----------------------------------------------------------------|
| 1 | Bulk student import via CSV                | S      | csv package present; onboarding 400 students by hand is brutal |
| 2 | Exam seating / invigilation roster planner | M      | rooms x classes x teachers; outputs printable PDFs             |
| 3 | Data-correction approval queue             | S      | receiving end of guardian correction requests                  |
| 4 | Document verification queue                | S      | receiving end of guardian document uploads                     |
| 5 | Timetable conflict detector                | S      | teacher double-booked / class with no teacher, flagged on save |

## Teacher

| #  | Feature                                                                  | Effort | Notes                                                                                      |
|----|--------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------|
| 1  | "My class today" morning card                                            | S      | aggregates absentees, birthdays, leave requests, duties, copies due — all existing queries |
| 2  | Live "now teaching" strip (current period + countdown)                   | S      | bell timings + timetable already stored                                                    |
| 3  | Leave balance view <b>[GAP+ — no balance concept in code]</b>            | S      | quotas in settings; ledger from approved leave_applications                                |
| 4  | Canned one-tap notes to a single guardian                                | S      | "forgot copy", "uniform incomplete" -> per-student push                                    |
| 5  | Private student notes (never shown to guardians)                         | S      | distinct from shared student_remarks                                                       |
| 6  | Report card comment bank                                                 | S      | category picker, name auto-fill; saves hours per term                                      |
| 7  | Marks entry speedups (auto-advance, max validation, CSV import)          | S      | marks deadline week is peak stress                                                         |
| 8  | Extend offline queue to marks + homework <b>[GAP+ — attendance-only]</b> | M      | generalize enqueue/syncAll pattern; prevents lost-marks disasters                          |
| 9  | Photo roster for substitutes                                             | S      | name + photo + roll for an unknown class; needs student photos                             |
| 10 | Period swap requests (teacher<->teacher, coordinator notified)           | M      | rides substitution infrastructure                                                          |
| 11 | Homework recycling (repeat last, templates, copy to other class)         | S      | pure QoL on HomeworkService                                                                |
| 12 | "My class" analytics (teacher-scoped charts)                             | S      | filtered reuse of analytics_screen widgets                                                 |
| 13 | Voice input / audio attachments for remarks & homework                   | M      | Hindi typing pain; speech-to-text or short audio clips                                     |
| 14 | Printable blank registers & mark sheets                                  | S      | pdf + printing + student data                                                              |
| 15 | Stationery / resource request to admin                                   | S      | chalk, markers, projector booking                                                          |

## Guardian

| # | Feature                                                                          | Effort | Notes                                                        |
|---|----------------------------------------------------------------------------------|--------|--------------------------------------------------------------|
| 1 | Downloadable fee receipts <b>[GAP+ — receipt numbers exist, no guardian UI]</b>  | S      | gapless numbering already in fee_service; 80C tax use case   |
| 2 | Holiday calendar view <b>[GAP+ — calendar_events exists, no guardian screen]</b> | S      | "is school open tomorrow?" — read-only view of existing data |
| 3 | Data correction requests (name spelling, phone, DOB)                             | S      | DPDP right-to-correction; coordinator approves               |

| #  | Feature                                                                       | Effort | Notes                                                     |
|----|-------------------------------------------------------------------------------|--------|-----------------------------------------------------------|
| 4  | Multi-year report card archive [ <b>GAP+ — no session dimension</b> ]         | S      | stamp session now or history is lost at promotion         |
| 5  | Second guardian access (invite flow)                                          | M      | father + mother + grandparent on one child                |
| 6  | Urgent alert tier (separate channel, bypasses silent)                         | S      | priority field exists but is text-only; rain-day closures |
| 7  | Exam syllabus per exam ("Ch 1–4")                                             | S      | one field on exam model                                   |
| 8  | Class-average context on marks                                                | S      | "is 65 good?" — anonymous aggregate per exam-subject      |
| 9  | Child's leaderboard standing [ <b>GAP+ — service exists, no guardian UI</b> ] | S      | engagement feature already computed, just unshown         |
| 10 | Ask-the-teacher query box (ticket-style, one open per child)                  | M      | structured to protect teacher evenings                    |
| 11 | Weekly child digest push (attendance, homework, dues)                         | S      | guardian-sized principal_digest                           |
| 12 | Term-start essentials page (booklist, uniform, stationery)                    | S      | cheapest feature on this list                             |
| 13 | Guardian-side document upload                                                 | S      | feeds document vault; saves a school visit                |
| 14 | Exam countdown reminders ("Maths in 3 days")                                  | S      | scheduled Cloud Function over exam dates                  |
| 15 | Certificate request flow (bonafide etc., with status)                         | S      | request side of the cert generators                       |
| 16 | Guardian timetable view (child's subjects per day)                            | S      | "pack the bag"; data in timetable collection              |

## Deferred / Phase 2

- **Multi-tenancy / multi-school** — already consciously deferred
- **GPS bus tracking** — phase 2 of transport lite
- **Payment gateway (Razorpay etc.)** — phase 2 of UPI deep-link
- **Student login role** — app currently has no student account; senior classes could use one
- **WhatsApp Business API** — official template messages instead of share-sheet
- **Hostel module** — only if a customer school needs it

## Top 10 (desperation / effort, across all roles)

1. **Fee receipts for guardians** — numbering exists, UI missing, yearly 80C demand
2. **Holiday calendar for guardians** — data exists, screen missing
3. **Absence auto-alert** — one Cloud Function, daily value for every parent
4. **Exam result analytics for principal** — confirmed gap, foundation for 3 more features
5. **Teacher "my class today" card + leave balance** — daily teacher retention
6. **Daily cash reconciliation for owner** — anti-fraud, mostly queries
7. **Expense tracking / P&L for owner** — transforms app's value proposition
8. **UPI fee payment** — biggest perceived value, no gateway cost
9. **Offline queue extension to marks** — prevents the app-abandoning disaster
10. **WhatsApp share on PDFs** — trivial, instantly visible to everyone

\*Strategic note: features that create a daily reason to open the app (morning card,

diary, digests, alerts) compound — they keep users engaged so every other feature gets seen. Prioritize at least one per role early.\*